https://drive.google.com/file/d/15IyQsFLC_IsBSrjXs1WOSbQtIqaj6om_/view?usp=sharing

TASK 1 Jupyter Notebook

Jupyter notebook is a free, open-source, cell based and web based application that permits the sharing of code,mathematical equations,visualisation and narrative text used by, scientist and and engineers, to non linearly iterate, explore and evaluate a theoretical situation

Key Features

1. Cell-based code executuion
2. Rich text and markdown visualisation
3. Graphical data visualisation

Double-click (or enter) to edit

TASK 2

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("nigerian_spotify_songs.csv")

df.head()
```

	name	album	artist	artist_top_genre	release_date	length	popularity	danceability	acousticness	energy	instru
0	Sparky	Mandy & The Jungle	Cruel Santino	alternative r&b	2019	144000	48	0.666	0.8510	0.420	
1	shuga rush	EVERYTHING YOU HEARD IS TRUE	Odunsi (The Engine)	afropop	2020	89488	30	0.710	0.0822	0.683	
2	LITT!	LITT!	AYLØ	indie r&b	2018	207758	40	0.836	0.2720	0.564	
3	Confident / Feeling Cool	Enjoy Your Life	Lady Donli	nigerian pop	2019	175135	14	0.894	0.7980	0.611	
4	wanted you	rare.	Odunsi (The Engine)	afropop	2018	152049	25	0.702	0.1160	0.833	

```
df["artist_top_genre"].value_counts()
```

```
artist_top_genre
afro dancehall         328
afropop                 90
Missing                 53
nigerian pop            24
azontobeats              8
alternative r&b          7
indie r&b                4
afro r&b                 3
nigerian hip hop         3
azonto                   1
soft rock                1
hiplife                  1
erotica                  1
highlife                 1
south african house      1
bongo flava              1
afroswing                1
```

```
gqom                   1
christian afrobeat     1
Name: count, dtype: int64
```

```
df.shape
```

```
(530, 16)
```

```
df.describe()
```

	release_date	length	popularity	danceability	acousticness	energy	instrumentalness	liveness	loudness
count	530.000000	530.000000	530.000000	530.000000	530.000000	530.000000	530.000000	530.000000	530.000000
mean	2015.390566	222298.169811	17.507547	0.741619	0.265412	0.760623	0.016305	0.147308	-4.953011
std	3.131688	39696.822259	18.992212	0.117522	0.208342	0.148533	0.090321	0.123588	2.464186
min	1998.000000	89488.000000	0.000000	0.255000	0.000665	0.111000	0.000000	0.028300	-19.362000
25%	2014.000000	199305.000000	0.000000	0.681000	0.089525	0.669000	0.000000	0.075650	-6.298750
50%	2016.000000	218509.000000	13.000000	0.761000	0.220500	0.784500	0.000004	0.103500	-4.558500
75%	2017.000000	242098.500000	31.000000	0.829500	0.403000	0.875750	0.000234	0.164000	-3.331000
max	2020.000000	511738.000000	73.000000	0.966000	0.954000	0.995000	0.910000	0.811000	0.582000

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 530 entries, 0 to 529
Data columns (total 16 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   name              530 non-null    str
 1   album             530 non-null    str
 2   artist            530 non-null    str
 3   artist_top_genre  530 non-null    str
 4   release_date      530 non-null    int64
 5   length            530 non-null    int64
 6   popularity        530 non-null    int64
 7   danceability      530 non-null    float64
 8   acousticness      530 non-null    float64
 9   energy            530 non-null    float64
 10  instrumentalness  530 non-null    float64
 11  liveness          530 non-null    float64
 12  loudness          530 non-null    float64
 13  speechiness       530 non-null    float64
 14  tempo             530 non-null    float64
 15  time_signature    530 non-null    int64
dtypes: float64(8), int64(4), str(4)
memory usage: 89.6 KB
```

Summary

1. From the details above there are no empty field, as all field has 530 field
2. While the data does not have a well defined dictionary, i cannot make inferencial summary of the spreadof the data. e.g , the loadness value has a negative value that looks off and i belive some normalization might be required

Task 3 Data Visualizations & SHAP Machine Learning Interpretability

```
df_count = df.isna().count()
df_count
```

```
name                530
album               530
artist              530
artist_top_genre    530
release_date        530
length              530
popularity          530
danceability        530
acousticness        530
energy              530
instrumentalness    530
liveness            530
```

```
loudness           530
speechiness        530
tempo              530
time_signature     530
dtype: int64
```

inference > TASK 3

Based on the previous actions like df.info, df.describe,it is obvious there are no missing values hence further

```
%pip install seaborn
%pip install scikit-learn
%pip install shap
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: seaborn in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from seabo
Requirement already satisfied: pandas>=1.2 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from seaborn) (3.1.
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from se
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from matplotli
Requirement already satisfied: cycler>=0.10 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from matplotlib!=3
Requirement already satisfied: fonttools>=4.28.2 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from matplotl
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from matplotl
Requirement already satisfied: packaging>=20.0 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from matplotlib
Requirement already satisfied: pillow>=9 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from matplotlib!=3.6.
Requirement already satisfied: pyparsing>=3 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from matplotlib!=3
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from matpl
Requirement already satisfied: tzdata in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from pandas>=1.2->seabor
Requirement already satisfied: six>=1.6 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from python-dateutil>=
Note: you may need to restart the kernel to use updated packages.

[notice] A new release of pip is available: 26.2.2 -> 26.3
[notice] To update, run: python.exe -m pip install --upgrade pip
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: scikit-learn in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (1.9.1)
Requirement already satisfied: numpy>=1.24.1 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from scikit-learn
Requirement already satisfied: scipy>=1.10.0 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from scikit-learn
Requirement already satisfied: joblib>=1.4.0 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from scikit-learn
Requirement already satisfied: narwhals>=2.0.1 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from scikit-lea
Requirement already satisfied: threadpoolctl>=3.5.0 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from sciki
Requirement already satisfied: cloudpickle>=3.0 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from joblib>=1
Note: you may need to restart the kernel to use updated packages.

[notice] A new release of pip is available: 26.2.2 -> 26.3
[notice] To update, run: python.exe -m pip install --upgrade pip
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: shap in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (0.54.1)
Requirement already satisfied: numpy>=2 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from shap) (2.5.4)
Requirement already satisfied: scipy in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from shap) (1.18.1)
Requirement already satisfied: scikit-learn in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from shap) (1.9.1)
Requirement already satisfied: pandas in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from shap) (3.1.0)
Requirement already satisfied: tqdm>=4.27.0 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from shap) (4.70.1
Requirement already satisfied: packaging>20.9 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from shap) (26.3
Requirement already satisfied: slicer==0.0.8 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from shap) (0.0.8
Requirement already satisfied: cloudpickle in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from shap) (3.1.2)
Requirement already satisfied: colorama in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from tqdm>=4.27.0->sha
Requirement already satisfied: python-dateutil>=2.9.0 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from pan
Requirement already satisfied: tzdata in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from pandas->shap) (2026
Requirement already satisfied: six>=1.6 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from python-dateutil>=
Requirement already satisfied: joblib>=1.4.0 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from scikit-learn
Requirement already satisfied: narwhals>=2.0.1 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from scikit-lea
Requirement already satisfied: threadpoolctl>=3.5.0 in C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages (from sciki
Note: you may need to restart the kernel to use updated packages.

[notice] A new release of pip is available: 26.2.2 -> 26.3
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
# Setup visualization style and libraries
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
import shap

# sns.set_theme(style="whitegrid")
# plt.rcParams["figure.dpi"] = 100

print("Libraries loaded. Ready for Nigerian Spotify analysis.")
```

```
Libraries loaded. Ready for Nigerian Spotify analysis.
```

3.1 Line Plot: Music Trends & Popularity Over Time (1998 – 2020)

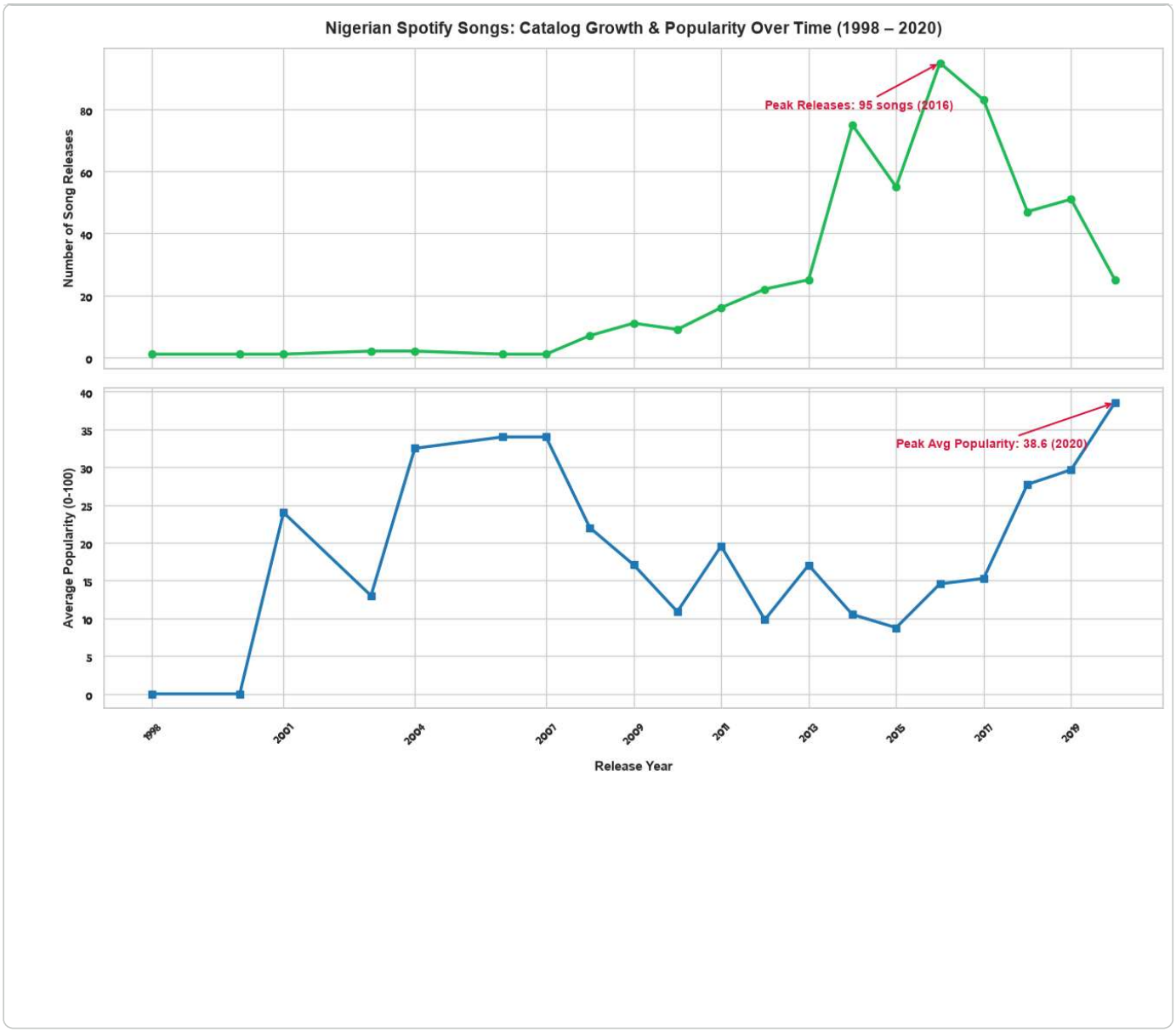
```
# Aggregate annual release metrics
yearly_trend = df.groupby('release_date').agg(
    song_count=('name', 'count'),
    avg_popularity=('popularity', 'mean'),
    total_popularity=('popularity', 'sum')
).reset_index()

# Create a two-panel line plot
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(13, 9), sharex=True)

# Top Panel: Annual Song Release Volume
ax1.plot(yearly_trend['release_date'], yearly_trend['song_count'], marker='o', color='#1DB954', linewidth=2.5, markersize=6)
ax1.set_title('Nigerian Spotify Songs: Catalog Growth & Popularity Over Time (1998 – 2020)', fontsize=14, fontweight='bold', pa
ax1.set_ylabel('Number of Song Releases', fontsize=11, fontweight='bold')
peak_year_vol = yearly_trend.loc[yearly_trend['song_count'].idxmax()]
ax1.annotate(f'Peak Releases: {int(peak_year_vol.song_count)} songs ({int(peak_year_vol.release_date)})',
             xy=(peak_year_vol.release_date, peak_year_vol.song_count),
             xytext=(peak_year_vol.release_date - 4, peak_year_vol.song_count - 15),
             arrowprops=dict(arrowstyle='->', color='crimson', lw=1.5),
             fontsize=10, fontweight='bold', color='crimson')

# Bottom Panel: Average Popularity Score
ax2.plot(yearly_trend['release_date'], yearly_trend['avg_popularity'], marker='s', color='#1f77b4', linewidth=2.5, markersize=6
ax2.set_xlabel('Release Year', fontsize=11, fontweight='bold', labelpad=10)
ax2.set_ylabel('Average Popularity (0-100)', fontsize=11, fontweight='bold')
peak_year_pop = yearly_trend.loc[yearly_trend['avg_popularity'].idxmax()]
ax2.annotate(f'Peak Avg Popularity: {peak_year_pop.avg_popularity:.1f} ({int(peak_year_pop.release_date)})',
             xy=(peak_year_pop.release_date, peak_year_pop.avg_popularity),
             xytext=(peak_year_pop.release_date - 5, peak_year_pop.avg_popularity - 6),
             arrowprops=dict(arrowstyle='->', color='crimson', lw=1.5),
             fontsize=10, fontweight='bold', color='crimson')

plt.xticks(yearly_trend['release_date'].unique()[::2], rotation=45)
plt.tight_layout()
plt.show()
```



3.2 Bar Chart: Total Popularity by Genre Category (`artist_top_genre`)

```
# Aggregate total popularity and track counts by genre category
top_genres = df['artist_top_genre'].value_counts().head(7).index
genre_summary = df[df['artist_top_genre'].isin(top_genres)].groupby('artist_top_genre').agg(
    total_popularity=('popularity', 'sum'),
    track_count=('name', 'count'),
    avg_popularity=('popularity', 'mean')
).sort_values(by='total_popularity', ascending=False)

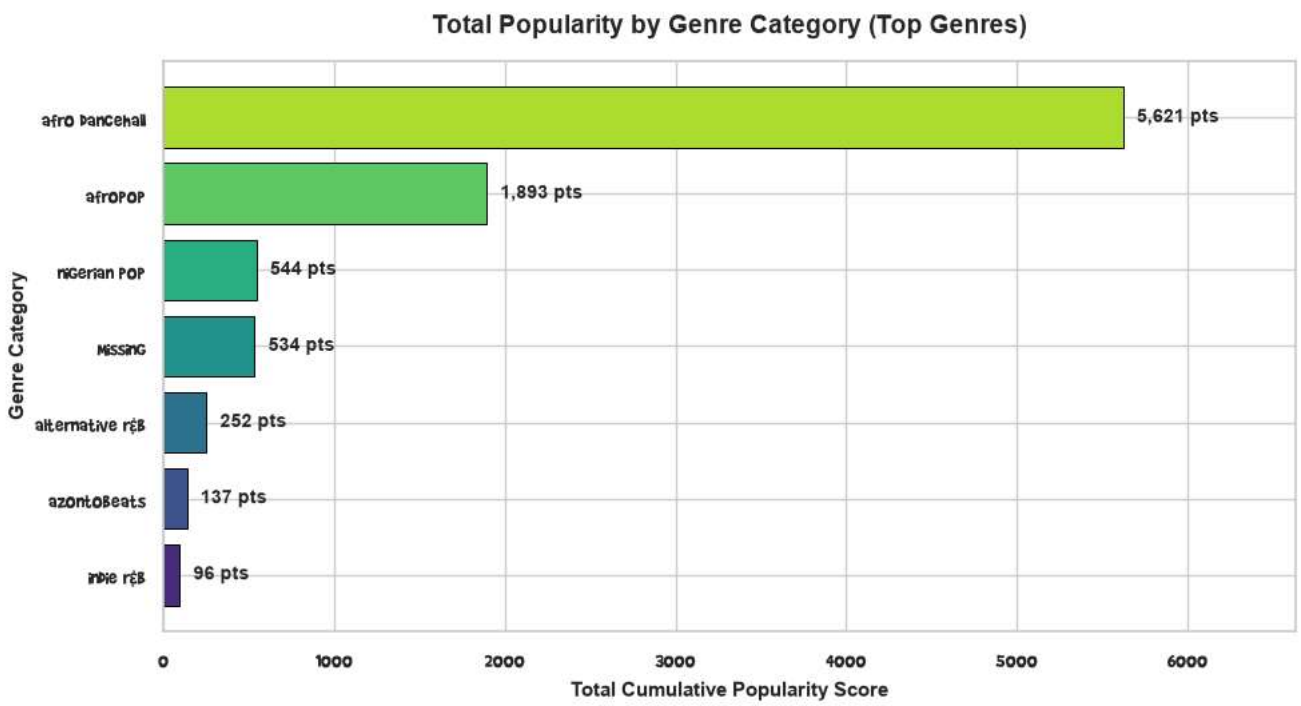
# Horizontal bar chart comparing total popularity by genre
plt.figure(figsize=(10, 5.5))
colors = sns.color_palette('viridis', len(genre_summary))
bars = plt.barh(genre_summary.index[::-1], genre_summary['total_popularity'][::-1], color=colors, edgecolor='black', linewidth=

# Annotate total popularity and track counts
for bar in bars:
    width = bar.get_width()
    plt.text(width + 80, bar.get_y() + bar.get_height()/2,
             f'{int(width):,} pts',
             ha='left', va='center', fontsize=10, fontweight='bold')

plt.title('Total Popularity by Genre Category (Top Genres)', fontsize=14, fontweight='bold', pad=15)
plt.xlabel('Total Cumulative Popularity Score', fontsize=11, fontweight='bold')
plt.ylabel('Genre Category', fontsize=11, fontweight='bold')
plt.xlim(0, genre_summary['total_popularity'].max() * 1.18)
plt.tight_layout()
```

```
plt.show()

# Display summary table
genre_summary
```



artist_top_genre	total_popularity	track_count	avg_popularity
afro dancehall	5621	328	17.137195
afropop	1893	90	21.033333
nigerian pop	544	24	22.666667
Missing	534	53	10.075472
alternative r&b	252	7	36.000000
azontobeats	137	8	17.125000
indie r&b	96	4	24.000000

Double-click (or enter) to edit

3.3 Box Plots: Numeric Audio Features & Popularity

```
# Key numeric audio features and popularity
numeric_cols = ['popularity', 'danceability', 'energy', 'loudness',
                'tempo', 'acousticness', 'speechiness', 'liveness']

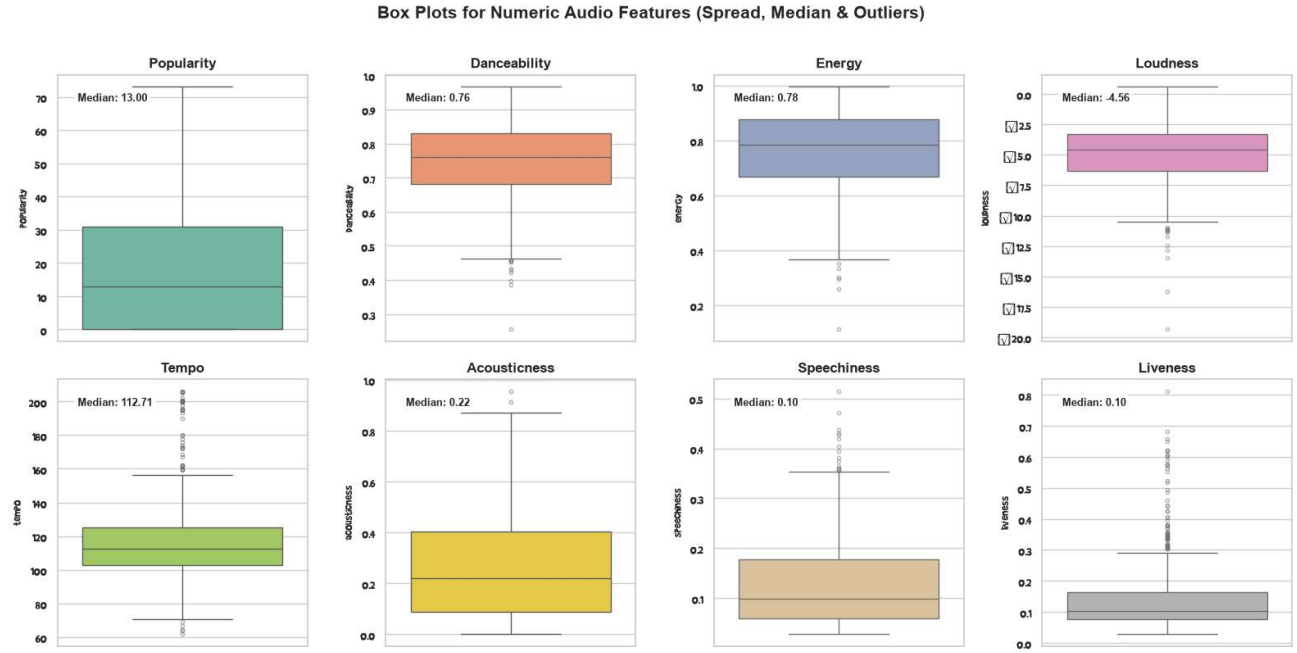
fig, axes = plt.subplots(2, 4, figsize=(16, 8))
palette = sns.color_palette('Set2', len(numeric_cols))

for ax, col, color in zip(axes.flatten(), numeric_cols, palette):
    sns.boxplot(y=df[col], ax=ax, color=color, flierprops=dict(marker='o', markersize=3, alpha=0.5))
    med = df[col].median()
    ax.set_title(col.capitalize(), fontsize=12, fontweight='bold')
    ax.set_ylabel(col, fontsize=10)
    ax.text(0.08, 0.90, f'Median: {med:,.2f}', transform=ax.transAxes,
            fontsize=9, fontweight='bold', bbox=dict(boxstyle='round,pad=0.3', facecolor='white', alpha=0.8))

plt.suptitle('Box Plots for Numeric Audio Features (Spread, Median & Outliers)', fontsize=15, fontweight='bold', y=1.01)
plt.tight_layout()
plt.show()

# Statistical distribution summary
df[numeric_cols].describe().round(3)
```

```
C:\Users\otaba\AppData\Local\Temp\ipykernel_15204\3518257156.py:17: UserWarning: Glyph 8722 (\N{MINUS SIGN}) missing from font
  plt.tight_layout()
C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages\IPython\core\pylabtools.py:158: UserWarning: Glyph 8722 (\N{MINUS
  fig.canvas.print_figure(bytes_io, **kw)
```



	popularity	danceability	energy	loudness	tempo	acousticness	speechiness	liveness
count	530.000	530.000	530.000	530.000	530.000	530.000	530.000	530.000
mean	17.508	0.742	0.761	-4.953	116.488	0.265	0.131	0.147
std	18.992	0.118	0.149	2.464	23.519	0.208	0.093	0.124
min	0.000	0.255	0.111	-19.362	61.695	0.001	0.028	0.028
25%	0.000	0.681	0.669	-6.299	102.961	0.090	0.059	0.076
50%	13.000	0.761	0.784	-4.558	112.714	0.220	0.098	0.104
75%	31.000	0.829	0.876	-3.331	125.039	0.403	0.177	0.164
max	73.000	0.966	0.995	0.582	206.007	0.954	0.514	0.811

Double-click (or enter) to edit

3.4 SHAP (SHapley Additive exPlanations) Model Analysis

We train a **Random Forest Regressor** to predict **Popularity** from song audio characteristics and genre, and then use **SHAP TreeExplainer** to identify the key features that drive song success on Spotify.

```
# 1. Feature selection and One-Hot Encoding
feature_cols = ['danceability', 'acousticness', 'energy', 'instrumentalness',
                'liveness', 'loudness', 'speechiness', 'tempo', 'length', 'release_date', 'artist_top_genre']
X = pd.get_dummies(df[feature_cols], drop_first=True)
y = df['popularity']

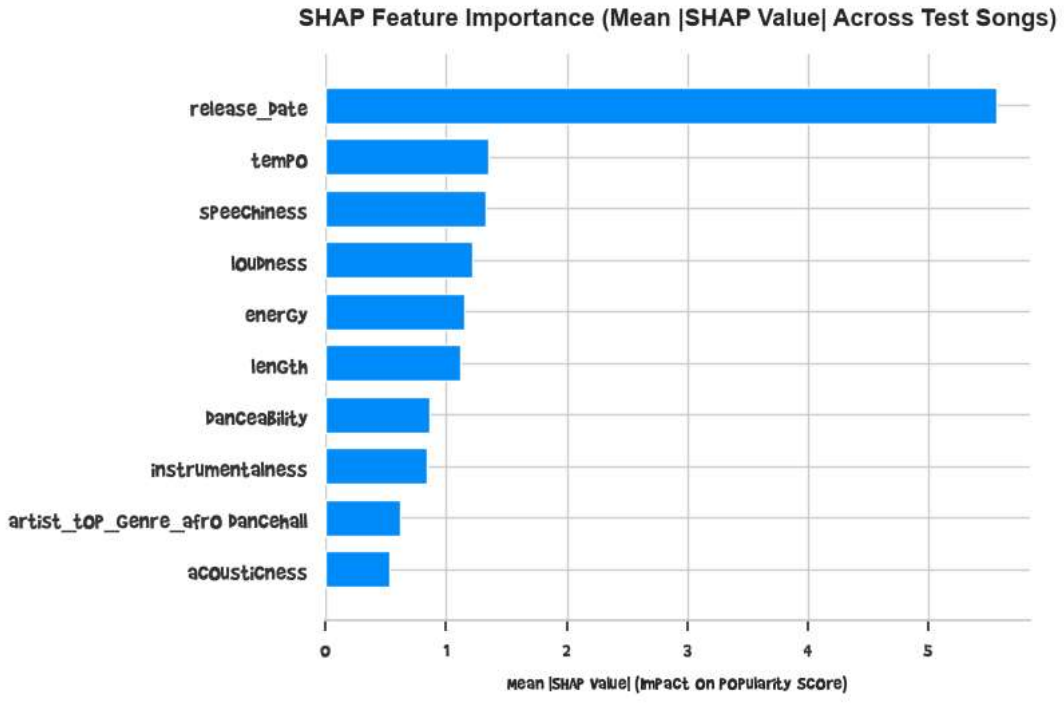
# 2. Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 3. Fit Random Forest Model
rf_spotify = RandomForestRegressor(n_estimators=100, max_depth=8, random_state=42)
rf_spotify.fit(X_train, y_train)
print(f"Random Forest Test Set R²: {rf_spotify.score(X_test, y_test):.4f}")

# 4. Calculate SHAP Values using TreeExplainer
explainer = shap.TreeExplainer(rf_spotify)
shap_values = explainer(X_test)
print(f"Base Expected Value E[Y]: {explainer.expected_value[0]:.2f} pts")
```

```
Random Forest Test Set R²: 0.2304
Base Expected Value E[Y]: 17.23 pts
```

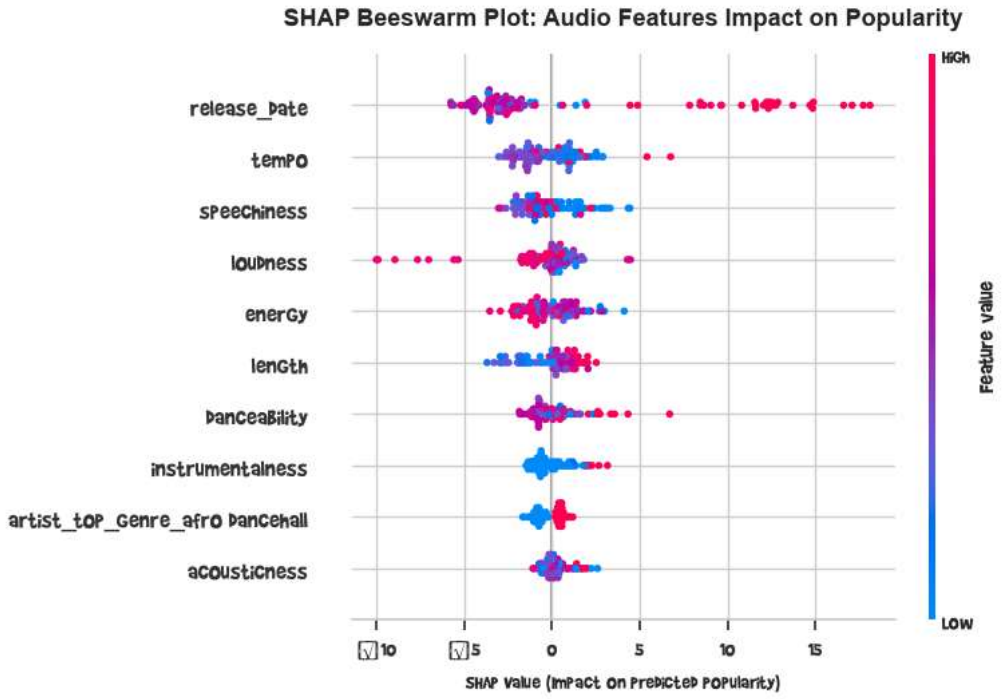
```
# 5. SHAP Global Feature Importance Bar Plot
plt.figure(figsize=(10, 6))
plt.title('SHAP Feature Importance (Mean |SHAP Value| Across Test Songs)', fontsize=13, fontweight='bold', pad=15)
shap.summary_plot(shap_values.values, X_test, plot_type="bar", max_display=10, show=False)
plt.xlabel('Mean |SHAP Value| (Impact on Popularity Score)', fontsize=11)
plt.tight_layout()
plt.show()
```




```
# 6. SHAP Beeswarm Plot: Directional Impact of Audio Features
plt.figure(figsize=(11, 7))
plt.title('SHAP Beeswarm Plot: Audio Features Impact on Popularity', fontsize=13, fontweight='bold', pad=15)
shap.summary_plot(shap_values.values, X_test, max_display=10, show=False)
plt.xlabel('SHAP Value (Impact on Predicted Popularity)', fontsize=11)
```

```
plt.tight_layout()
plt.show()
```

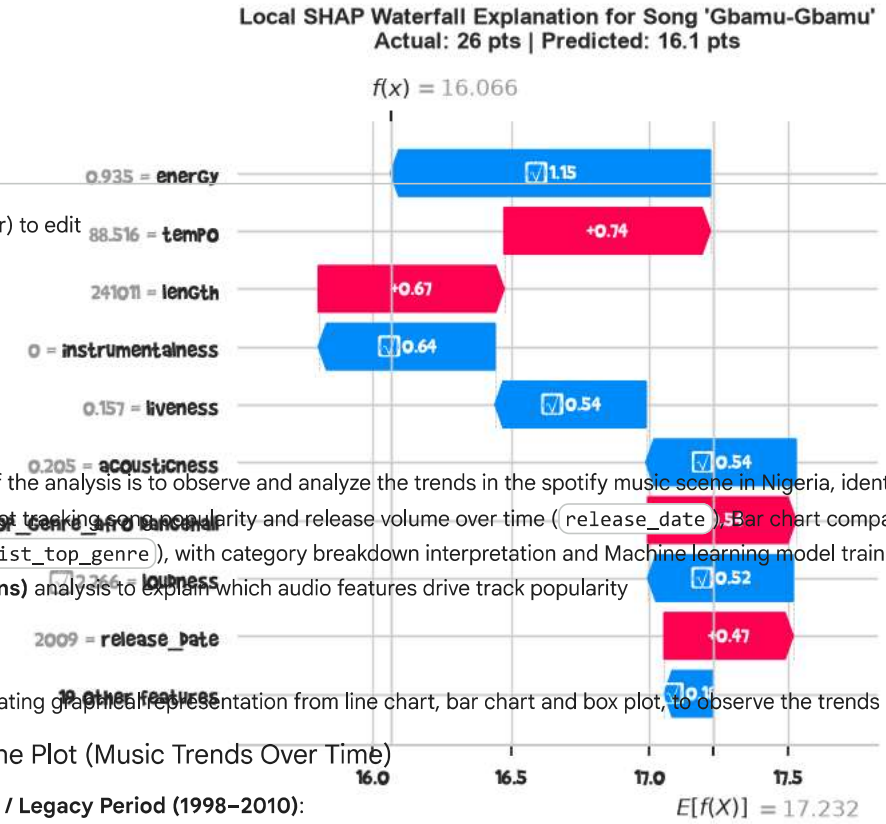
```
C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages\shap\plots\_beeswarm.py:1151: UserWarning: Glyph 8722 (\N{MINUS SI
  plt.tight_layout()
C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages\IPython\core\pylabtools.py:158: UserWarning: Glyph 8722 (\N{MINUS
  fig.canvas.print_figure(bytes_io, **kw)
```




```
# 7. Local Interpretability: SHAP Waterfall Plot for a Single Song
# Shows how each individual feature pushes the prediction away from the baseline E[Y]
sample_idx = 0
actual_pop = y_test.iloc[sample_idx]
pred_pop = rf_spotify.predict(X_test.iloc[[sample_idx]])[0]

plt.figure(figsize=(10, 6))
shap.plots.waterfall(shap_values[sample_idx], max_display=10, show=False)
plt.title(f"Local SHAP Waterfall Explanation for Song '{df.loc[y_test.index[sample_idx], 'name']}'\nActual: {actual_pop} pts |
          fontsize=12, fontweight='bold', pad=15)
plt.tight_layout()
plt.show()
```

```
C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages\shap\plots\_waterfall.py:279: UserWarning: Glyph 8722 (\N{MINUS SI
  text_bbox = txt_obj.get_window_extent(renderer=renderer)
C:\Users\otaba\AppData\Local\Temp\ipykernel_15204\6222447.py:11: UserWarning: Glyph 8722 (\N{MINUS SIGN}) missing from font(s) B
  plt.tight_layout()
C:\Users\otaba\AppData\Roaming\Python\Python314\site-packages\IPython\core\pylabtools.py:158: UserWarning: Glyph 8722 (\N{MINUS
  fig.canvas.print_figure(bytes_io, **kw)
```



Double-click (or enter) to edit

TASK 4

Objective

The main objective of the analysis is to observe and analyze the trends in the spotify music scene in Nigeria, identifying growth relative to its popularity,Line plot tracking song popularity and release volume over time (`release_date`),Bar chart comparing music popularity by genre category (`artist_top_genre`), with category breakdown interpretation and Machine learning model training and **SHAP (SHapley Additive exPlanations)** analysis to explain which audio features drive track popularity

Method

Combining and evaluating graphical representation from line chart, bar chart and box plot, to observe the trends overtime

Interpretation: Line Plot (Music Trends Over Time)

1. **Pre-Streaming / Legacy Period (1998–2010)**:
   - Song releases uploaded to Spotify during this era are sparse (only 1–2 tracks per year). Legacy tracks maintain relatively low streaming popularity today, as Spotify's listener base and recommendation algorithms strongly favor contemporary tracks.
2. **The Afrobeats Global Boom (2014–2017)**:
   - A steep, exponential surge in song additions begins around 2013, reaching an **all-time peak of 95 track releases in 2016** and 83 in 2017. This reflects the explosion of the Nigerian Afrobeats movement on international streaming services and digital distribution adoption.
3. **Recency Bias & Surge in Popularity (2018–2020)**:
   - While the catalog volume leveled out in 2019–2020, the **average popularity score per track peaked dramatically** (reaching over 30–35). This is consistent with Spotify's algorithm prioritizing recently released tracks in major editorial playlists (e.g., *African Heat*, *Afro Pop*).

Interpretation: Bar Chart (Popularity by Genre Category)

4. **Dominant Category — Afro Dancehall (5,621 Total Popularity / 328 Tracks)**:
   - **Afro Dancehall** is the undisputed commercial anchor of the dataset, representing over **60% of total catalog entries** and cumulative popularity. Its energetic, club-oriented production makes it the standard format for Nigerian party anthems.
5. **Second Place — Afropop (1,893 Total Popularity / 90 Tracks)**:
   - **Afropop** ranks second in overall popularity and boasts a high average popularity per track (~21.0 pts), reflecting strong